

RETAIL CUSTOMER ANALYTICS & SEGMENTATION

RFM Analysis • K-Means Clustering • Power BI • Marketing Strategy

PROJECT FOCUS

Transforming retail transaction data into customer-level insights, actionable segments, and targeted marketing strategies.

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Project Report

Public retail transaction dataset | RFM & customer segmentation business case

Table of Contents

1. Executive Summary
2. Introduction
3. Problem Statement
4. Project Objectives
5. Scope of the Project
6. Dataset Description
7. Data Preparation and Cleaning
8. Exploratory Business Context
9. RFM Analysis
10. Data Transformation and Clustering Methodology
11. K-Means Evaluation and Cluster Selection
12. Final Customer Segmentation
13. Power BI Dashboard
14. Key Business Findings
15. Segment-Wise Marketing Strategy
16. Managerial Implications
17. Limitations
18. Future Scope
19. Conclusion
20. Technical Implementation
21. References
22. Appendix

Note: Page numbers can be refreshed in Microsoft Word after opening the document if an automatic Table of Contents is added.

1. Executive Summary

This project develops an end-to-end retail customer analytics solution using a public Online Retail II transaction dataset. The objective is to move from transaction-level records to customer-level business insight by combining data cleaning, RFM (Recency, Frequency, Monetary) analysis, K-Means clustering, and Power BI visualization.

The source data contained 1,067,371 transaction rows across two sheets. After removing records without customer identifiers, cancellations/returns, non-positive quantities or prices, and duplicates, 779,425 usable transaction rows remained. The cleaned data represented 5,878 customers, 36,969 invoices, and £17.37 million in transaction value.

RFM analysis quantified customer behavior using recency, purchase frequency, and monetary value. The RFM variables were log-transformed and standardized before K-Means clustering. Cluster counts from K=2 through K=8 were evaluated using inertia and silhouette score. Although K=2 produced the highest silhouette score of 0.4386, K=4 was selected because it provided more differentiated and actionable customer groups for marketing decision-making.

KEY RESULT

20.35% of customers were classified as Champions and contributed approximately 73.87% of total revenue. This demonstrates a strong concentration of revenue among a relatively small high-value customer group.

The final solution translates the analytical output into four business-oriented segments: Champions, At-Risk Valuable, New / Promising, and Low-Engagement / Lost. A five-page Power BI dashboard was built to communicate customer behavior, segment economics, transaction/product patterns, and recommended marketing actions.

2. Introduction

Retail organizations generate large volumes of transaction data through customer purchases. However, transaction records alone do not directly answer important managerial questions such as which customers generate the greatest value, which customers are becoming inactive, and which customers should receive retention, win-back, or development campaigns.

Customer segmentation provides a structured way to group customers according to shared characteristics or behaviors. In this project, segmentation is behavior-based: customers are grouped according to their historical purchasing activity rather than demographic assumptions.

RFM analysis is used as the foundation because it captures three practical dimensions of customer value: how recently a customer purchased, how frequently they purchased, and how much monetary value they generated. K-Means clustering is then applied to identify groups with similar RFM profiles.

The final stage is business translation. Rather than stopping at a machine-learning output, the project maps each segment to a marketing objective and a set of practical actions.

3. Problem Statement

How can customer transaction data be used to identify distinct customer segments and develop targeted marketing strategies for each segment?

The problem is therefore both analytical and managerial: identify meaningful differences in customer behavior and convert those differences into decisions around retention, reactivation, customer development, and marketing resource allocation.

4. Project Objectives

- Clean and prepare raw retail transaction data for customer-level analysis.
- Calculate customer-level Recency, Frequency, and Monetary metrics.
- Identify behavioral customer segments using K-Means clustering.
- Evaluate alternative cluster counts using inertia and silhouette score.
- Interpret the final segments from a marketing and business perspective.
- Develop a five-page Power BI dashboard for interactive analysis and communication.
- Translate segment-level findings into targeted marketing recommendations.

5. Scope of the Project

The project covers:

- Historical transaction-level retail data.
- Customer identification through Customer ID.
- Transaction cleaning and value calculation.
- RFM-based customer behavior analysis.
- Unsupervised customer segmentation using K-Means.
- Business interpretation of segment characteristics.
- Power BI dashboarding and marketing recommendations.

The project does not cover:

- Real-time customer behavior.
- Proprietary data from Reliance Trends or another specific retailer.
- Causal measurement of campaign effectiveness.
- A/B testing of the proposed marketing strategies.
- Predictive churn or purchase-propensity modeling.

6. Dataset Description

The project uses the public Online Retail II dataset. The source workbook contains two sheets covering successive retail periods. The available fields include Invoice, StockCode, Description, Quantity, InvoiceDate, Price, Customer ID, and Country.

Field	Description / Role
Invoice	Invoice or transaction identifier; cancellation invoices are

	identifiable by a leading C.
StockCode	Product identifier.
Description	Product description.
Quantity	Quantity purchased in the transaction.
InvoiceDate	Date and time of the transaction.
Price	Unit price.
Customer ID	Customer identifier used for customer-level analysis.
Country	Customer country / market.

The raw dataset contained 1,067,371 rows. The analysis used only transaction records that could be attributed to a customer and that met the defined transaction-quality rules.

DATA PROVENANCE NOTE

The dataset is public. It is not proprietary Reliance Trends data. The project may be discussed through a retail/fashion marketing lens, but no proprietary Reliance data or confidential company information is claimed.

7. Data Preparation and Cleaning

Data preparation was performed in Python using Pandas. The cleaning process was implemented as a reproducible script (`data\_cleaning.py`) so that the processed dataset can be regenerated from the source workbook.

7.1 Cleaning Procedure

1. Combine the two source sheets into a single transaction table.
2. Remove rows without Customer ID because customer-level segmentation requires an identifiable customer.
3. Remove cancellation/return invoices beginning with the letter C.
4. Keep only Quantity > 0.
5. Keep only Price > 0.
6. Remove duplicate records.
7. Create Transaction_Value = Quantity × Price.
8. Export the cleaned transaction data to CSV for downstream analysis.

7.2 Cleaning Results

Metric	Result
Original transaction rows	1,067,371
Clean transaction rows	779,425
Rows removed	287,946
Customers	5,878
Invoices	36,969
Total transaction value	£17,374,804.27

The cleaning stage therefore reduced the working dataset while preserving the customer-linked positive-value transactions required for RFM analysis.

8. Exploratory Business Context

The cleaned dataset provides the basis for understanding customer purchasing activity at an aggregate level. The analysis contains 5,878 identifiable customers and 36,969 invoices after cleaning.

The main business purpose of this stage is not to produce a large number of descriptive charts, but to establish a reliable transaction base from which customer behavior can be measured. The subsequent RFM analysis shifts the unit of analysis from individual transaction rows to individual customers.

This distinction is important: a customer may appear across many transaction rows and invoices. Customer-level aggregation therefore provides a more useful foundation for retention and value-based marketing decisions.

9. RFM Analysis

RFM analysis summarizes customer purchasing behavior using three dimensions.

Metric	Business Meaning	Interpretation
Recency	How recently the customer purchased.	Lower values indicate more recent activity.
Frequency	How often the customer purchased.	Higher values indicate more repeat purchasing.
Monetary	How much transaction value the customer generated.	Higher values indicate greater historical customer value.

9.1 RFM Calculation

The snapshot date was defined as one day after the maximum InvoiceDate in the cleaned transaction data. The resulting snapshot date was 10 December 2011 at 12:50:00.

For each customer, Recency was calculated from the most recent purchase date to the snapshot date. Frequency was measured as the number of distinct invoices associated with the customer. Monetary value was calculated as the sum of Transaction_Value.

9.2 RFM Results

Metric	Mean	Median / Maximum where available
Recency	201.33 days	Median 96 days; Maximum 739 days
Frequency	6.29 orders	Median 3; Maximum 398
Monetary	£2,955.90	Median £867.74; Maximum £580,987.04

The gap between mean and median monetary value indicates a highly uneven distribution of customer value, with a smaller number of customers generating substantially higher transaction value. This is one reason why segmentation is useful for marketing resource allocation.

10. Data Transformation and Clustering Methodology

K-Means clustering is sensitive to scale and can be influenced by highly skewed variables. RFM variables, especially Monetary and Frequency, commonly contain large values and long tails. The project therefore used a two-stage preprocessing approach.

9. Apply log1p transformation to Recency, Frequency, and Monetary values to reduce the influence of extreme values.
10. Standardize the transformed variables using StandardScaler so that the dimensions are placed on a comparable scale.

The transformed and standardized RFM matrix was then used as the input to K-Means. A fixed random state of 42 and n_init=10 were used in the implementation to improve reproducibility.

10.1 Why K-Means?

K-Means was selected because the project aims to discover groups of customers with similar behavioral profiles without predefined labels. It is also relatively easy to interpret and translate into segment-level business actions.

11. K-Means Evaluation and Cluster Selection

Cluster counts from K=2 through K=8 were tested. Two evaluation measures were retained: inertia and silhouette score.

Inertia measures within-cluster compactness; lower values generally improve as additional clusters are added. Silhouette score evaluates how well observations fit within their assigned cluster relative to neighboring clusters.

K	Inertia	Silhouette Score
2	8,588.99	0.4386
3	6,354.34	0.3477
4	4,921.23	0.3650
5	4,099.11	0.3425
6	3,554.70	0.3348
7	3,194.50	0.3066
8	2,902.43	0.3033

CLUSTER SELECTION DECISION

K=2 produced the highest silhouette score (0.4386). However, K=4 was selected for the final business segmentation because it produced four differentiated and actionable customer groups while maintaining a reasonable silhouette score of 0.3650.

This is a deliberate business-interpretability decision rather than a claim that K=4 is mathematically optimal on silhouette score alone. The final model is intended to support differentiated marketing actions.

12. Final Customer Segmentation

The final K=4 model produced four customer groups. Segment names were assigned by interpreting each cluster's average Recency, Frequency, Monetary value, customer share, and revenue contribution.

Segment	Customers	Customer Share	Avg. Recency	Avg. Frequency	Avg. Monetary	Revenue Share
Champions	1,196	20.35%	27.71 days	19.28	£10,731.16	73.87%
At-Risk	1,459	24.82%	230.07 days	5.06	£1,948.50	16.36%

Valuable						
New / Promising	1,250	21.27%	28.30 days	3.05	£857.49	6.17%
Low-Engagement / Lost	1,973	33.57%	394.95 days	1.38	£317.08	3.60%

12.1 Champions

Champions have very recent activity, high purchase frequency, and substantially higher monetary value. They represent 20.35% of customers but account for 73.87% of revenue.

12.2 At-Risk Valuable

This group has materially higher recency than Champions while retaining meaningful historical frequency and monetary value. They represent an important reactivation opportunity.

12.3 New / Promising

These customers are recent but have lower purchase frequency and monetary value than Champions. Their recency indicates current engagement, while their lower frequency indicates an opportunity to build repeat behavior.

12.4 Low-Engagement / Lost

This is the largest group by customer count, representing 33.57% of customers, but it contributes only 3.60% of revenue. The group has the highest average recency and lowest frequency and monetary value among the four segments.

13. Power BI Dashboard

A five-page Power BI dashboard was developed to translate the analytical output into an interactive business intelligence format. The dashboard is designed around a light, professional visual system with clear KPI cards, charts, segment comparisons, and strategy-oriented commentary.

Page 1 — Executive Overview

Provides the high-level business view through customer, revenue, and order KPIs; revenue contribution by segment; and customer distribution by segment. The main executive takeaway is the concentration of revenue among Champions.

Page 2 — Customer Behaviour & RFM Analysis

Examines average Recency, Frequency, and Customer Value by segment and uses a Frequency-versus-Monetary relationship view to make behavioral differences visible.

Page 3 — Customer Segmentation

Focuses on segment composition, revenue contribution, customer share, segment profiles, and average customer value.

Page 4 — Marketing Strategy & Recommendations

Connects the analytical segments to actions. Champions receive retention/value-expansion strategies; At-Risk Valuable customers receive win-back strategies; New / Promising customers receive repeat-purchase development actions; and Low-Engagement / Lost customers receive selective low-cost reactivation.

Page 5 — Transaction & Product Analysis

Provides transaction-level business context through revenue trends, country-level revenue, top products by revenue, and quantity-versus-transaction-value analysis.

14. Key Business Findings

Revenue concentration: 20.35% of customers (Champions) contribute 73.87% of total revenue.

Largest customer group: Low-Engagement / Lost customers represent 33.57% of the customer base but only 3.60% of revenue.

Retention opportunity: At-Risk Valuable customers contribute 16.36% of revenue and have materially higher historical value than Low-Engagement / Lost customers, making them a priority for win-back activity.

Growth opportunity: New / Promising customers show recent activity but lower frequency, creating an opportunity to encourage a second and subsequent purchase.

Uneven customer value: The monetary distribution is highly skewed: mean monetary value is £2,955.90 compared with a median of £867.74.

Segment differentiation: The final four segments show clear differences in recency, frequency, monetary value, and revenue contribution.

15. Segment-Wise Marketing Strategy

Segment	Primary Objective	Recommended Actions	Business Rationale
Champions	Retain and increase value	VIP rewards; early access; personalized recommendations; exclusive offers; cross-sell	High-value customers have the strongest demonstrated contribution and should be protected from churn while increasing relationship value.
At-Risk Valuable	Reactivate	Win-back campaigns; targeted offers; personalized recommendations; remarketing; limited-time reactivation	Historical value combined with inactivity makes this group a high-priority reactivation pool.
New / Promising	Build repeat behavior	Second-purchase offers; loyalty enrollment; product discovery; post-purchase engagement	Recent customers can be developed into more frequent customers.
Low-Engagement / Lost	Selectively reactivate	Low-cost automated campaigns; incentive tests; prioritize responsive customers	Large customer count but low revenue contribution suggests that expensive one-to-one retention efforts may be inefficient.

15.1 Strategic Principle

The central marketing implication is that customers should not be treated as one homogeneous audience. The same campaign objective and incentive structure is unlikely to be equally appropriate for a high-value Champion and a Low-Engagement / Lost customer.

16. Managerial Implications

- Prioritize retention resources around high-value customers because a relatively small customer group contributes a large share of revenue.
- Create a dedicated win-back journey for valuable inactive customers rather than treating all inactive customers equally.
- Use post-purchase engagement to increase frequency among New / Promising customers.
- Use cost-efficient automation and controlled incentive testing for Low-Engagement / Lost customers.
- Use Power BI as a management interface for monitoring segment composition, customer value, and revenue contribution.
- Treat the segments as decision-support groups that can be refreshed as new transaction data becomes available.

17. Limitations

- The dataset is historical and therefore does not represent current retail behavior.
- The source is a public dataset and is not proprietary data from Reliance Trends or another specific retailer.
- Segmentation is based primarily on historical transaction behavior and does not include demographic or attitudinal variables.
- K-Means is sensitive to preprocessing choices and the selected number of clusters.
- K=4 was selected for business actionability even though K=2 achieved the highest silhouette score.
- The proposed marketing strategies are analytical recommendations, not experimentally validated campaign results.
- The analysis does not establish causal relationships between customer characteristics and future revenue.

18. Future Scope

- Customer Lifetime Value (CLV) modeling to estimate future economic value.
- Cohort and retention analysis to understand customer lifecycle behavior.
- Predictive churn modeling to identify customers likely to become inactive.
- Purchase-propensity modeling for campaign targeting.
- Recommendation systems for personalized product discovery.
- A/B testing to validate the proposed segment-specific marketing interventions.
- Automated data refresh and deployment of the dashboard as an ongoing business intelligence workflow.

19. Conclusion

This project demonstrates an end-to-end approach to customer analytics in which technical analysis is connected directly to business decision-making. Starting with more than one million raw transaction rows, the workflow produced a clean customer-linked dataset, calculated RFM measures, evaluated multiple K-Means configurations, and developed four actionable customer segments.

The most significant finding is that approximately one-fifth of customers contribute nearly three-quarters of revenue. This reinforces the importance of differentiated customer strategies: retain high-value customers, reactivate valuable inactive customers, develop promising recent customers, and use cost-efficient approaches for low-engagement customers.

The Power BI dashboard turns these findings into an accessible management view, while the Python scripts make the analytical workflow reproducible. Together, the components demonstrate the use of data analytics not merely to describe customers, but to support practical marketing and strategic decisions.

20. Technical Implementation

20.1 Python Scripts

Script	Purpose
data_cleaning.py	Combines source sheets, applies cleaning rules, calculates Transaction_Value, and exports cleaned transactions.
rfm_analysis.py	Aggregates transactions at customer level and calculates Recency, Frequency, and Monetary value.
clustering.py	Transforms and standardizes RFM data, evaluates K=2–8, performs K-Means with K=4, and assigns business segment names.

20.2 Main Output Files

Output	Purpose
clean_retail_transactions.csv	Cleaned transaction-level dataset.
rfm.csv	Customer-level RFM dataset.
final_customer_segments.csv	Customer-level RFM data with final K-Means cluster and business segment.
Customer Sementation and RFM.pbix	Power BI dashboard file.

20.3 Reproducibility

The project is maintained as a Git repository with a README, Python scripts, processed data outputs, Power BI dashboard, and dashboard screenshots. The Python workflow can be rerun from the source dataset using the documented scripts.

21. References

- Online Retail II public retail transaction dataset (used as the analytical source dataset).
- Project-generated Python analysis outputs: clean_retail_transactions.csv, rfm.csv, final_customer_segments.csv.
- Project-generated Power BI dashboard: Customer Sementation and RFM.pbix.

Note: This report intentionally does not attribute any project result to proprietary retailer data. All quantitative results reported here are derived from the public transaction dataset and the project's documented analysis pipeline.

22. Appendix

Appendix A — Final Segment Metrics

Segment	Customers	Avg Recency	Avg Frequency	Avg Monetary	Revenue Share
Champions	1,196	27.71	19.28	£10,731.16	73.87%
At-Risk Valuable	1,459	230.07	5.06	£1,948.50	16.36%
New / Promising	1,250	28.30	3.05	£857.49	6.17%
Low-Engagement / Lost	1,973	394.95	1.38	£317.08	3.60%

Appendix B — K-Means Evaluation

K	Inertia	Silhouette
2	8588.99	0.4386
3	6354.34	0.3477
4	4921.23	0.3650
5	4099.11	0.3425
6	3554.70	0.3348
7	3194.50	0.3066
8	2902.43	0.3033

Appendix C — Project Repository

GitHub repository: [Customer-Segmentation-RFM](#)

Repository owner: Yash Patil

The repository contains the project README, Python scripts, processed datasets, Power BI dashboard, and screenshots.